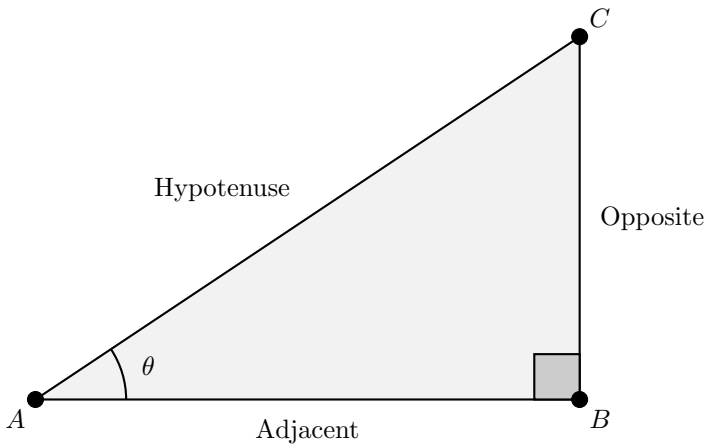
1 Right Triangles and Trigonometric Ratios

1.1 Similarity and Ratio Invariance

1.2 Definitions of the Six Ratios



Let $\triangle ABC$ be a right triangle with $\angle B = 90°$, and let $\theta = \angle BAC$ denote the angle at vertex A. Thus, the hypotenuse is the side AC, and the segment AB is adjacent and BC opposite to θ. The trigonometric functions are defined as the following:

Definition 1 (Sine)**.**

$$\sin\theta = \frac{BC}{AC} = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

Definition 2 (Cosecant)**.**

$$\csc\theta = \frac{AC}{BC} = \frac{\text{Hypotenuse}}{\text{Opposite}}$$

Definition 3 (Cosine)**.**

$$\cos\theta = \frac{AB}{AC} = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

Definition 4 (Secant)**.**

$$\sec\theta = \frac{AC}{AB} = \frac{\text{Hypotenuse}}{\text{Adjacent}}$$

Definition 5 (Tangent)**.**

$$\tan\theta = \frac{BC}{AB} = \frac{\text{Opposite}}{\text{Adjacent}}$$

Definition 6 (Cotangent)**.**

$$\cot\theta = \frac{AB}{BC} = \frac{\text{Adjacent}}{\text{Opposite}}$$

1.3 Bounds and Monotonicity

2 Basic Identities and Pythagorean Relations

2.1 Reciprocal Identities

Identity 1.

$$\sin\theta = \frac{1}{\csc\theta}, \quad \csc\theta = \frac{1}{\sin\theta}$$

Identity 2.

$$\cos\theta = \frac{1}{\sec\theta}, \quad \sec\theta = \frac{1}{\cos\theta}$$

Identity 3.

$$\tan\theta = \frac{1}{\cot\theta}, \quad \cot\theta = \frac{1}{\tan\theta}$$

2.2 Quotient Identities

Identity 4 (Tangent Identity)**.**

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

Identity 5 (Cotangent Identity)**.**

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

2.3 Pythagorean Theorem

Theorem 1 (Pythagorean Theorem)**.** *For right triangle $\triangle ABC$, the lengths of the sides satisfy:*

$$AB^2 + BC^2 = AC^2$$

2.4 Pythagorean Identity

Identity 6 (Pythagorean Identity)**.**

$$\sin^2\theta + \cos^2\theta = 1$$

Proof. Using the side-length definitions of sine and cosine for $\triangle ABC$:

$$\sin\theta = \frac{BC}{AC} \quad \text{and} \quad \cos\theta = \frac{AB}{AC}$$

Squaring both terms gives:

$$\sin^2\theta = \frac{BC^2}{AC^2} \quad \text{and} \quad \cos^2\theta = \frac{AB^2}{AC^2}$$

Adding them together:

$$\sin^2\theta + \cos^2\theta = \frac{BC^2}{AC^2} + \frac{AB^2}{AC^2} = \frac{AB^2 + BC^2}{AC^2}$$

By the Pythagorean Theorem, $AB^2 + BC^2 = AC^2$, which yields:

$$\sin^2\theta + \cos^2\theta = \frac{AC^2}{AC^2} = 1$$

□

2.5 Alternative Pythagorean Identities

Identity 7 (Pythagorean Identity)**.**

$$\sec^2\theta - \tan^2\theta = 1$$

Proof. From the Pythagorean Identity, divide both sides by $\cos^2\theta$, yielding

$$\begin{aligned} \sin^2\theta + \cos^2\theta = 1 &\implies \frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta} \\ &\implies \tan^2\theta + 1 = \sec^2\theta \\ &\implies \sec^2\theta - \tan^2\theta = 1 \end{aligned}$$

□

Identity 8 (Pythagorean Identity)**.**

$$\csc^2\theta - \cot^2\theta = 1$$

Proof. From the Pythagorean Identity, divide both sides by $\sin^2\theta$, yielding

$$\begin{aligned}\sin^2\theta + \cos^2\theta = 1 &\implies \frac{\sin^2\theta}{\sin^2\theta} + \frac{\cos^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta} \\ &\implies 1 + \cot^2\theta = \csc^2\theta \\ &\implies \csc^2\theta - \cot^2\theta = 1\end{aligned}$$

□

3 Cofunction Relationships and Complementary Angles

3.1 Complementary Angles

Definition 7 (Complementary Angles)**.** Two angles are *complementary* if and only if the sum of their measures is equal to the measure of a right angle (90° or $\frac{\pi}{2}$ radians).

Formally, let α and β be the measures of two angles. They are complementary if and only if $\alpha + \beta = 90°$. Either angle is referred to as the complement of the other.

Theorem 2 (Congruence of Complements)**.** *If two angles are complementary to the same angle, then they are congruent to each other. Similarly, if they are complementary to two congruent angles, they are congruent.*

Proof. Let angle α and angle β be complementary to a common angle γ.

By the definition of complementary angles, $\alpha + \gamma = 90°$ and $\beta + \gamma = 90°$.

By the transitive property of equality, we have

$$\alpha + \gamma = \beta + \gamma.$$

Subtracting γ from both sides yields

$$\alpha = \beta.$$

Thus, $\alpha \cong \beta$. □

Theorem 3 (Complementarity of Acute Angles in a Right Triangle)**.** *The two acute angles of any right-angled triangle are complementary.*

Proof. Let $\triangle ABC$ be a right triangle with $\angle B = 90°$. The sum of the interior angles of any triangle is 180°. Thus,

$$\angle A + \angle B + \angle C = 180°.$$

Substituting $\angle B = 90°$, we have

$$m(\angle A) + 90° + m(\angle C) = 180°$$

Subtracting 90° from both sides gives

$$m(\angle A) + m(\angle C) = 90°.$$

By definition, since the sum of $m(\angle A)$ and $m(\angle C)$ equals 90°, $\angle A$ and $\angle C$ are complementary. □

3.2 Cofunction Identities

Identity 9 (Cofunction Identities)**.**

$$\sin(90^\circ - \theta) = \cos\theta, \quad \cos(90^\circ - \theta) = \sin\theta$$

Proof. proof □

4 Extension to General Triangles

4.1 Law of Sines

Theorem 4 (Law of Sines)**.** *For any triangle $\triangle ABC$ with side lengths a, b, c opposite to angles α, β, γ respectively, the ratio of the sine of an angle to its opposite side is constant and equal to the reciprocal of the circumradius diameter:*

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma} = 2R$$

where R is the radius of the circumcircle of $\triangle ABC$.

4.2 Law of Cosines

Theorem 5 (Law of Cosines)**.** *For any triangle $\triangle ABC$ with side lengths a, b, c and corresponding opposite angles A, B, C:*

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = a^2 + c^2 - 2ac\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

4.3 Trigonometric Area Formula

5 Circle Geometry and Ptolemy's Theorem

5.1 Inscribed Angles and Cyclic Polygons

5.2 Ptolemy's Theorem

5.3 Angle Sum and Difference Formulas

Identity 10.

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Identity 11.

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

Identity 12.

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$

6 Multi-Angle Constructions

6.1 Double-Angle Identities

6.2 Power-Reducing Identities

6.3 Half-Angle Identities

6.4 Product and Sum Conversion Lemmas

7 Classical Chord Theory

7.1 Geometry of Chords and the Chord Function

7.2 Equivalence of Chords and Modern Ratios

7.3 Construction of Ancient Trigonometric Tables

8 Triangle Centers and Cevians

8.1 Trigonometric Ceva's Theorem

8.2 Trigonometric Menelaus's Theorem

8.3 Classical Centers

9 Advanced Metric Relations

9.1 Carnot's Theorem

9.2 Euler's Distance Formula

9.3 Pedal Triangles and Simson Lines